# Matthew Tian

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## EDUCATION

**Wilfrid Laurier University — Waterloo, ON** *Sep 2023 – May 2028*

*B.Sc. Computer Science & B.B.A. Economics (Double Degree) | GPA: 3.9/4.0* *3rd Year*

## CERTIFICATIONS

**Google CyberSecurity Certificate** *Online Course*
*Completed the online certification provided by Google* *May 2025 - July 2025*

- Completed Google's professional training program covering network security, threat detection, incident response, risk assessment, and security best practices.
- Gained hands-on experience with tools and concepts such as Linux command-line, SQL for security analysis, and frameworks like NIST Cybersecurity Framework.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, Java, C++, C, Swift, ARM Assembly, SQL
**ML & AI:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, LangChain, Claude API, Computer Vision APIs, NumPy, Pandas
**Web & Backend:** React.js, Node.js, Flask, REST APIs, HTML5, CSS3, AWS (S3, CloudFront, Lambda)
**Tools & Infra:** Git/GitHub, Docker, Linux, MySQL, Jupyter, Google Colab, NIST Framework, SSL/TLS

## PROJECTS

**NLP Sentiment Analyzer** *Python, PyTorch, Hugging Face | 2026*

- Fine-tuned a pre-trained BERT model on a labeled financial news dataset, achieving **91%** test accuracy; built a full preprocessing pipeline (tokenization, padding, batching) that reduced training time by 30%.
- Evaluated model performance with precision, recall, and F1-score across three sentiment classes; applied class-weight balancing to resolve label imbalance and improve minority-class recall by 14%.
- Currently in progress with AWS to deploy a trained model through Flask REST API to integrate low latency ping.

**Deep Convolutional GAN (DCGAN)** *Python, PyTorch | 2026*

- Implemented a DCGAN from scratch to synthesize 64x64 images; applied batch normalization and LeakyReLU activations across Generator/Discriminator networks to stabilize adversarial training dynamics.
- Preprocessed and augmented a 500+ image dataset via normalization, random flip, and center cropping; monitored training convergence using Frechet Inception Distance (FID) across epochs.
- Iteratively tuned learning rates, batch sizes, and architecture depth to eliminate mode collapse, producing visually coherent synthetic samples and demonstrating strong unsupervised learning fundamentals.

**Market Trend Classifier** *Python, scikit-learn, PostgreSQL | 2026*

- Fused TF-IDF headline embeddings with 15+ engineered OHLCV features (RSI, MACD, rolling volatility) in a Gradient Boosted ensemble, improving directional prediction F1 from 0.61 to 0.74 via k-fold cross-validation and grid-search tuning.
- Designed a recurring ETL pipeline in Python with psycopg2 to ingest, clean, and store structured financial data in PostgreSQL — directly paralleling distributed data workflows used in production ML systems.
- Wrote modular, documented code across data ingestion, feature engineering, training, and evaluation stages, making the pipeline reproducible and easy to extend with new data sources.

## EXPERIENCE

**Verdanza Tech — Enactus Laurier** *Sep 2025 – Present*
*AI Full Stack Developer* Waterloo, ON

- Built a web platform for local schools and grocery retailers integrating Computer Vision API for automated expiration decision across 500+ food products, eliminating manual inspection for routine checks
- Designed MySQL database schema to store product images with metadata (purchase date, expiration, quality scores), enabling automated quality assessment pipelines
- Developed multi-page React.js application with React Router, implementing lazy loading and optimized component rendering to improve page load performance
- Deployed frontend to AWS S3 with CloudFront CDN for distribution, built to support 1,000+ monthly active users

## LEADERSHIP

**WLU Big Backs — President** *Sep 2025 – Present*
*Wilfrid Laurier University Student Club*Waterloo, ON

- Oversee all club operations including event planning, member recruitment, and budget allocation for a growing student organization at Laurier.
- Coordinate with university administration and external partners to organize cross-campus events and community initiatives which grew Club membership by 40% within the first year